Amphibian Survey of South East Lois Hole Centennial Provincial Park



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BLESS
Big Lake Environment Support Society

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Executive Summary and Acknowledgments

Five amphibian species inhabit Lois Hole Centennial Provincial Park (LHCPP) yet little is known about their life history or status within park boundaries. Given the encroachment of surrounding developments, informal trail activity, and possible presence of at-risk species, this study aims to: 1)Confirm species presence 2)Identify required habitat conditions 3)Identify conflicts and recommend mitigation. A combination of overnight passive acoustic monitoring and day-time citizen-led efforts were carried out in May. Only passive monitoring was successful and confirmed the presence of Wood Frogs, Boreal Chorus Frogs, and Boreal Toads. Both frog species preferred riparian habitat, utilizing submerged cattails to support egg masses. Boreal Toad choruses were detected further inland, likely in shallow vernal pools and water-filled depressions. Boreal Toads are of concern as breeding habitat is limited here. Closures of informal trails, amendments to the current trail system, increased parks presence,and new signs are needed to mitigate visitor impact. Further monitoring is needed to identify post-breeding movements and other breeding groups in and around the park.

I would like to thank Miles Constable, Ciara Fraser, and Deena Larsen for their help with project permits, equipment procurement, citizen science procedure design, and volunteer advertising. I would also like to thank the Alberta Conservation Association for their amazing amphibian identification guide and Kris Kendell for his help in identifying Boreal Toad calls.

Introduction

Bordering the cities of St. Albert and Edmonton is Lois Hole Centennial Provincial Park (LHCPP) encompassing 1,357 hectares of protected wetlands, forests, and the waters of Big Lake[8]. The southeast parcel is the only accessible part of LHCPP and is home to mixed-wood forests dominated by trembling aspen and balsam poplar, with fragments of white spruce[9].

Eight trails have been established for hikers and cyclists, including a wetland interpretive boardwalk. However, multiple unofficial trails have been recorded over the years. These trails likely start off as game trails, made by moose and deer, but are mistaken by visitors for footpaths. Repeated use of these trails disturbs emergent vegetation and causes soil-compaction, which lowers water percolation, plant reestablishment, and recovery.
Unofficial trail use is high during winter months, where foot-traffic compresses snow, crushing underlying vegetation. By spring, these trails are visible and indiscernible from the official trail system. Flooding brought on by annual snowmelt flows lakewards, collecting on trails. To avoid waters, visitors either stick to trail edges — damaging vegetation and widening trails — or choose detours via unofficial trails, causing permanent damage over time.

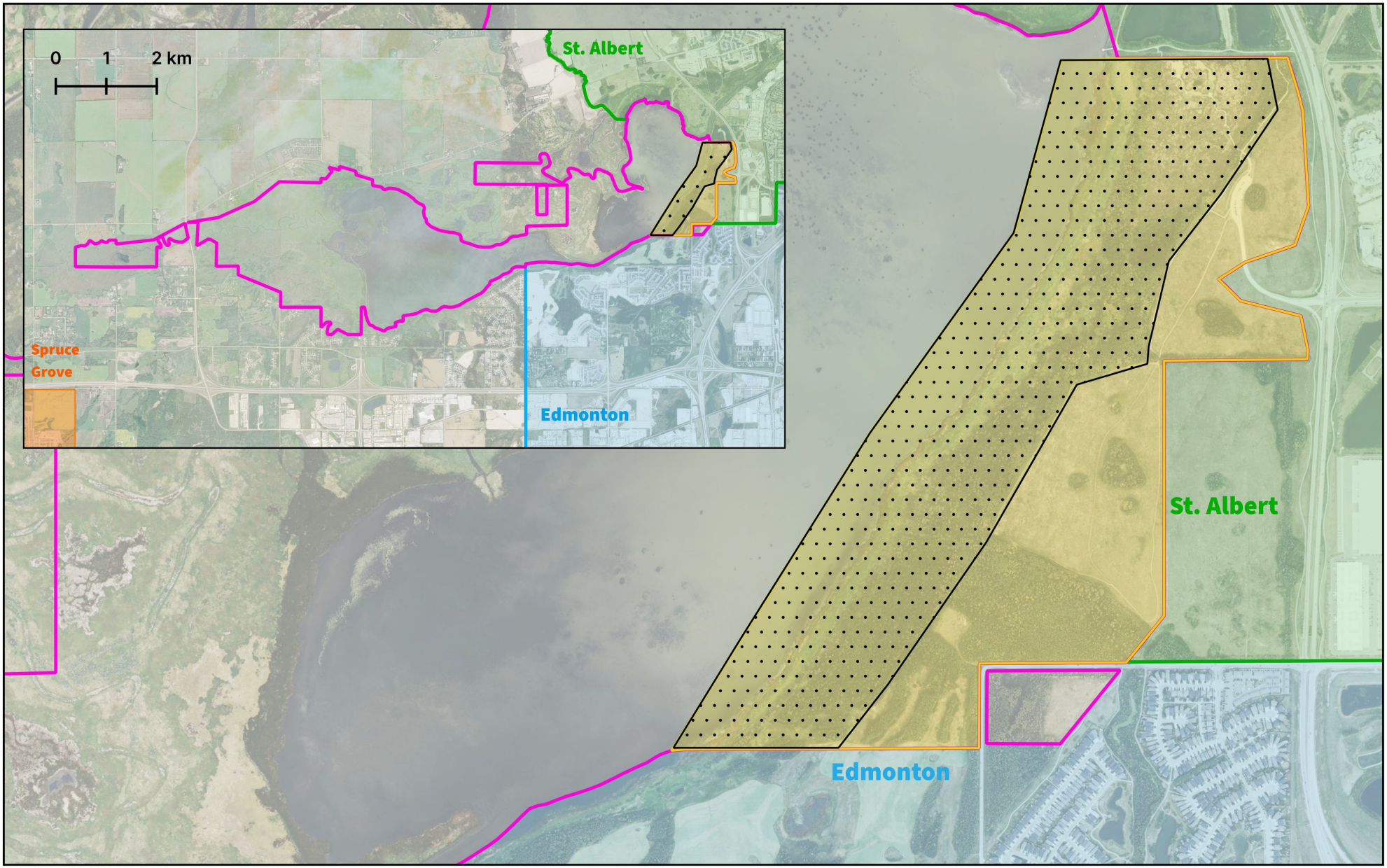
While this area serves as a corridor for large ungulates and numerous migratory birds, it is also home to a small selection of amphibians [4]. Species include: Wood Frog (*Lithobates sylvaticus*), Boreal Chorus Frog(*Pseudacris maculata*), Barred Tiger Salamander(*Ambystoma mavortium*), Boreal Toad(*Anaxyrus boreas*), and Canadian Toad(*Anaxyrus hemiophrys*). These species likely gather in snowmelt-recharged vernal pools, wetlands, and flooded riparian zones to reproduce. Local amphibians serve two important ecological roles: As predators, they control invertebrate populations; as prey, they serve as nutrient-rich meals for resident species like the Great Blue Heron and Northern Pike[12].
While observations are made, not much is known about amphibians within LHCPP. This is especially true of Boreal Toads, which are recognized as “Sensitive” in Alberta[13]. Boreal Toads are particularly sensitive to habitat degradation and amphibian Chytrid Fungus [1].While searching for salamanders requires more invasive techniques, frog and toad presence can be determined passively by deploying audio recorders. Furthermore, Alberta frog and toad calls are distinctive and can be easily identified by nature enthusiasts, opening the possibility for continuous monitoring.
As impacts on local amphibians remain unknown, persistent visitor disturbance continues while more existential risks from new developments and infrastructure continue to encroach on important habitats.

Given the risks to local amphibians, this monitoring program was established with the following objectives in mind:

- Confirm which anurans occur within study area
- Identify required habitat conditions within LHCPP that support each species
- Identify human-anuran conflicts and recommend mitigation where necessary

Lois Hole Centennial Provincial Park

Study Area



Study Area
Southeast Parcel
Park Boundary
Edmonton
St. Albert
Spruce Grove

0 250 500 m

Methods

Overnight Monitoring

Three recorders were used to monitor overnight mating calls. Each device consisted of: 1 wooden stake, 1 metal rain cover with a sponge fixed to the top, and one *V15-Pro voice recorder* attached below the cover via zip-ties. Each recorder was labeled with an identification number, project permit number, and contact information(appendix 3). Monitoring was conducted from May 4th to May 29th. The recorders were established at designated sites, paralleling Moose trail(Figure 2). Each recorder was placed and activated at approximately 20:00 MST and were left to record until 06:00 MST the following day when they were retrieved. Device settings where standardized(table below).

Setting	Value
Gain	5
Bit Rate	160 Kbps
Sampling Rate	48KHz
Recording Length	1 hr
Noise Reduction	0

Data Analysis

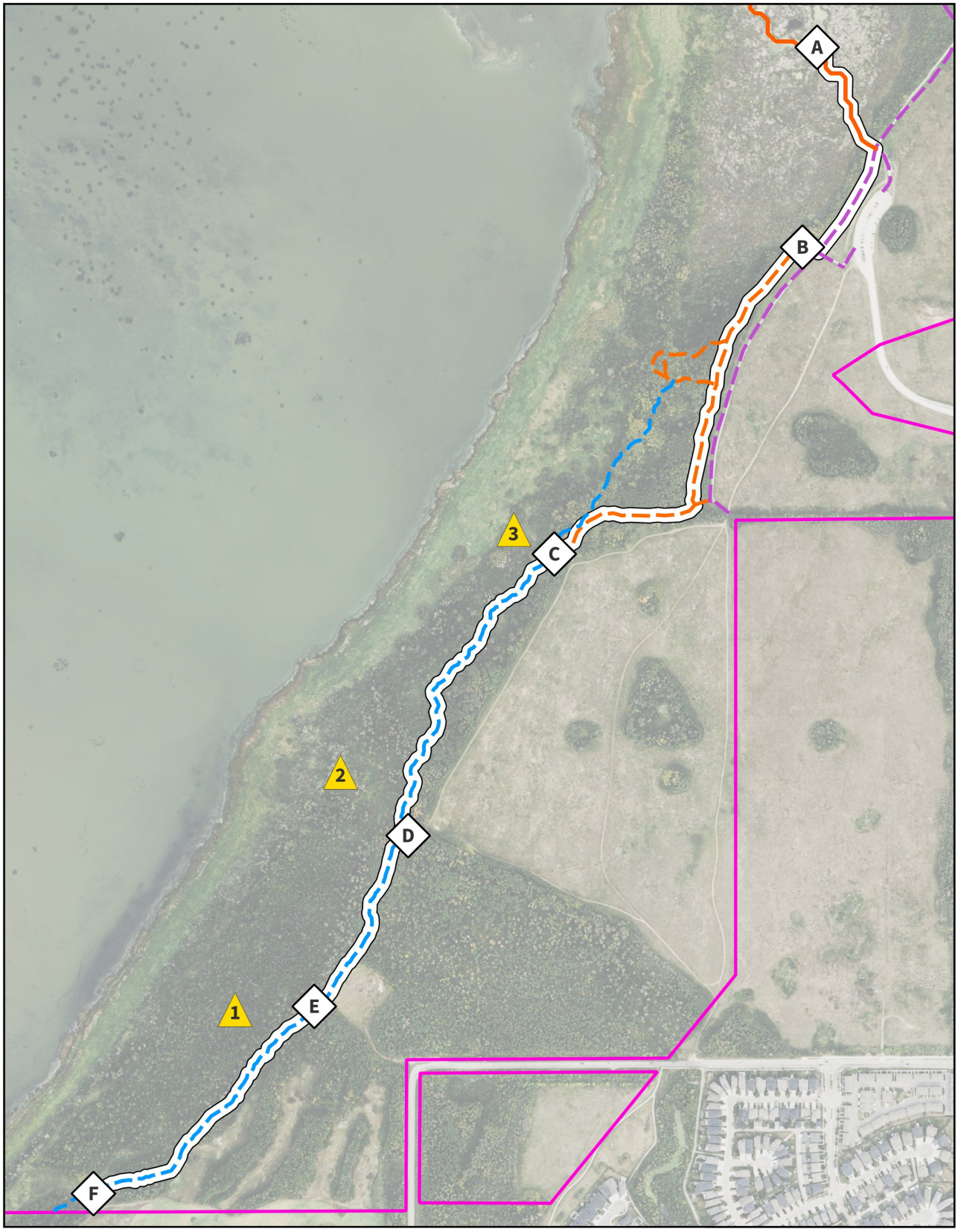
Only recordings between 21:00 and 05:00 MST were selected for analysis, a set of recordings within this time frame represents one “recording night”(appendix 1). Timing of each recording hour deviated by ±10 minutes between sites, but is compensated for during tabulation. Each recorded hour was given an index per species based on calling intensity(table below). This was determined by listening to each recording and examining its spectrogram via the open-source tool “Sonic-Visualizer”. The programming language R was used to compile and graphically display calling intensity over time, both code and data files are available via GitHub(appendix 2).

Call Index	Description	Spectrogram
0	No calls	No signatures
1	Calls not overlapping	Signatures visible; gaps between calls large
2	Calls overlapping	Signatures visible; gaps between calls small, forming continuous bands

Daytime Monitoring

This component involved local nature enthusiasts to ensure continuous daytime observation. Volunteers were instructed to walk a pre-selected trail, stopping at multiple points to listen and look for amphibians(figure 2). At each site, volunteers were to listen for 5 minutes for any frogs, then classify calling intensity by calling index (table above). Once completed, volunteers looked for anurans at the site or on the trail between the next site, species name and number of individuals were recorded.

Lois Hole Centennial Provincial Park Survey Paths and Positions



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Volunteer Path
Microphone
Volunteer Survey Points
Poplar Trail
Moose Trail
Coyote Trail
Park Boundary
Boardwalk
0 100 200 m

Results

Due to low daytime monitoring participation, no data was collected and has been excluded from this report. Weekend acoustic monitoring was avoided to prevent equipment theft or encouragement of informal trail use. Overnight monitoring of site 2 was terminated due to a malfunctioning recorder. Other recording periods were canceled due to weather and wildlife concerns(appendix 1).

Site 1 was set in a riparian area, with depths reaching 40 cm. Cattails were pressed down, 5-6 cm beneath the surface. Thicker cattail growths separated the site from the lake. Several egg masses were found on the path to site 1, belonging to either L. sylvaticus or P. maculata. To minimize disturbance, a designated path to the site was used and equipment was cleaned before and after use.
Site 3 was set in a white spruce stand, vernal pools here were not large and continuous like site 1, but were limited to deep, dispersed depressions. Each depression was approximately 15-30 cm deep and 0.5 m in diameter, each hosting numerous mosquito larvae later in the month. These depressions are likely the result of uprooted wind-thrown trees, snags that have undergone complete decomposition, a combination of the two, or possibly animal activity. The pools are likely recharged every spring by snowmelt and supplemented through summer months by rainfall. Larger pools likely exist closer to the lake.

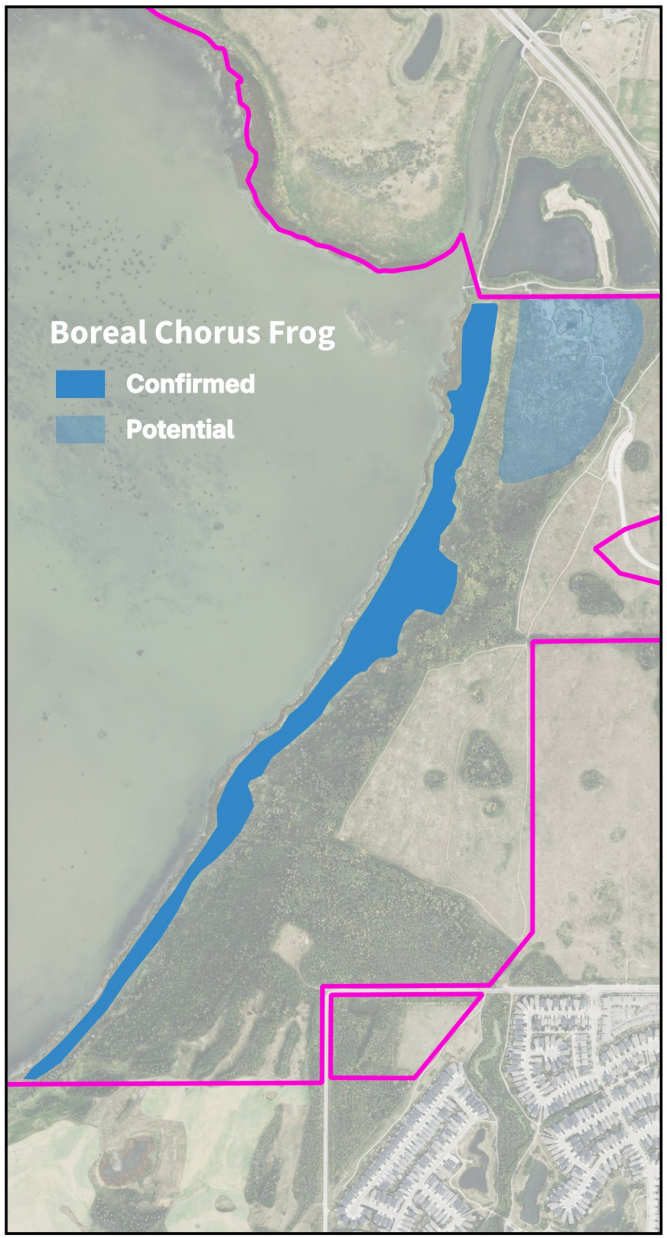
P. maculata was the most active anuran detected: Strong, multi-hour choruses were recorded at site 1. Despite frequency lowering in late May, choruses were still detected(figure 4). Site 3 presence was mixed; limited and faint call overlapping did occur but was likely 2-3 distant males north of the recorder(figure 5).
L. sylvaticus was another very active species. Multi-hour chorus continued at site 1 until night 7, only to subside over May(figure 6). This is consistent with their 1-2 week “explosive” breeding behavior[2]. Site 3 saw limited activity, either from lone males or distant L. sylvaticus calling groups(figure 7). Distant L. sylvaticus calling groups were heard at both sites, suggesting that breeding activity occurs throughout the Southeast riparian zone.

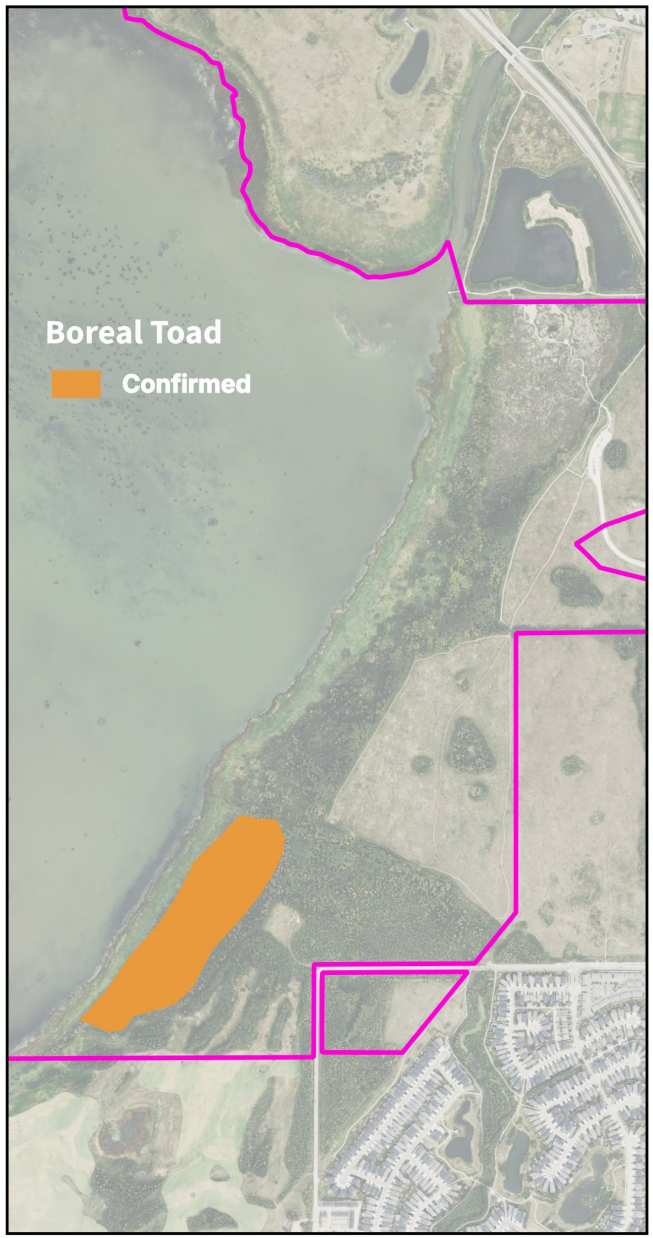
A. boreas was the only toad species detected. While both sites showed Boreal Toad presence, only site 3 produced multi-hour choruses(Figures 8-9). They were more distinct compared to other anuran signatures in site 3, showing that A. boreas prefer more inland vernal pools.

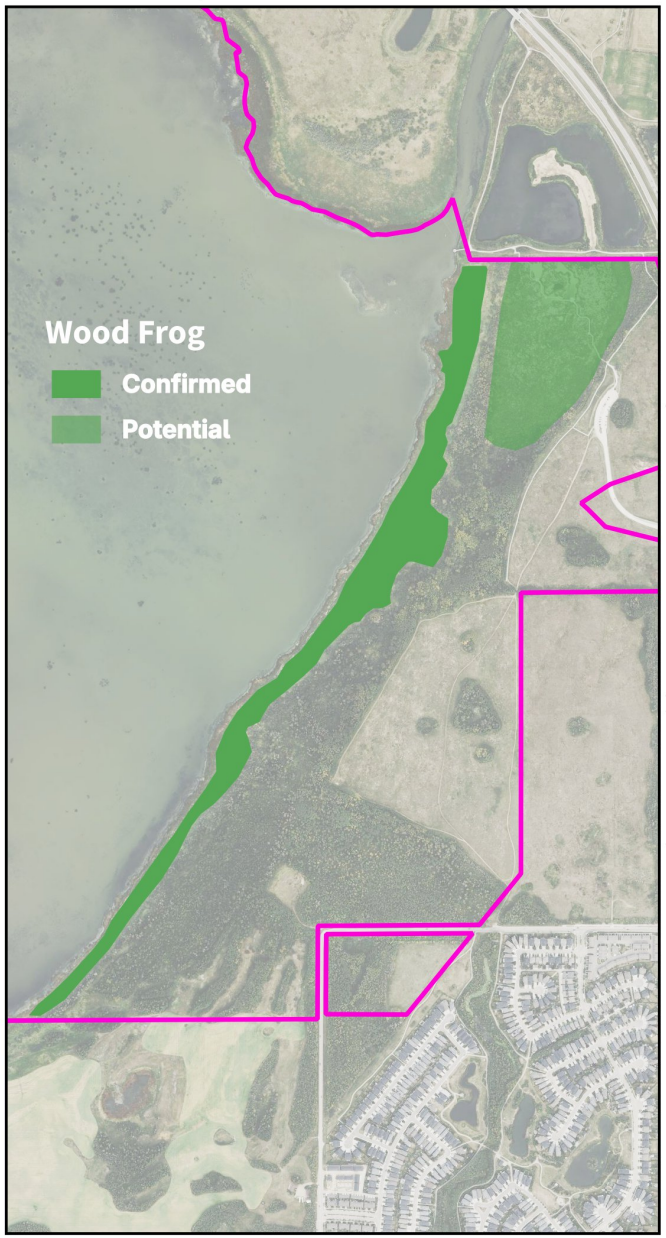
At 2.5-3.5 kHz, P. maculata calls occupied the highest frequency bands. A. boreas calling frequencies were low, ranging from 1-1.7 kHz. L. sylvaticus utilized the widest frequency range at 3.5-1.7 kHz. L. sylvaticus was the most vocally-dominant species and could out-compete P. maculata and A. boreas choruses if breeding periods conflict. However, it appears that A. boreas has compensated for this by beginning choruses after peak L. sylvaticus activity and in more remote vernal pools, whereas P. maculata choruses persist, capitalizing on weaker L. sylvaticus calls later in May and likely into June.

Lois Hole Centennial Provincial Park

Species Breeding Habitat





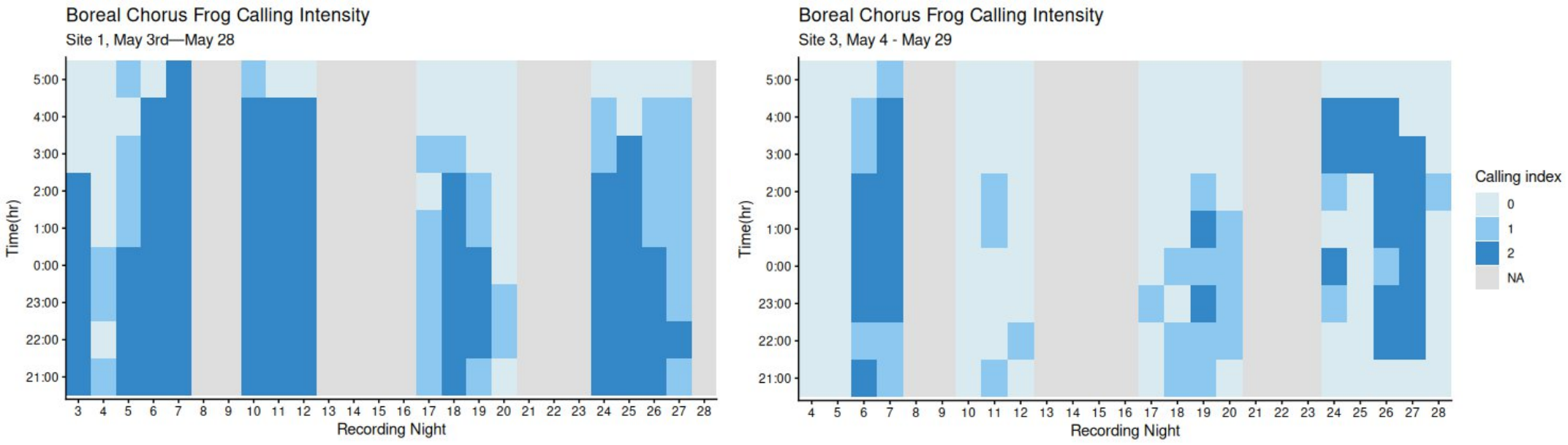


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Park Boundary

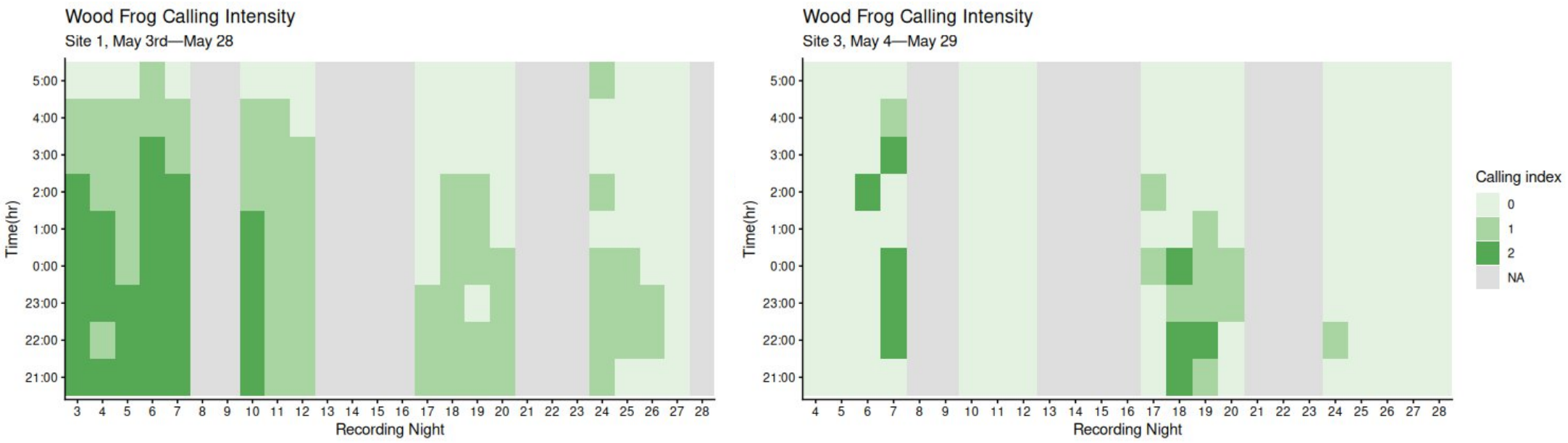
0 250 500 m

Figures 4-5 – Boreal Chorus Frog Calling Intensity



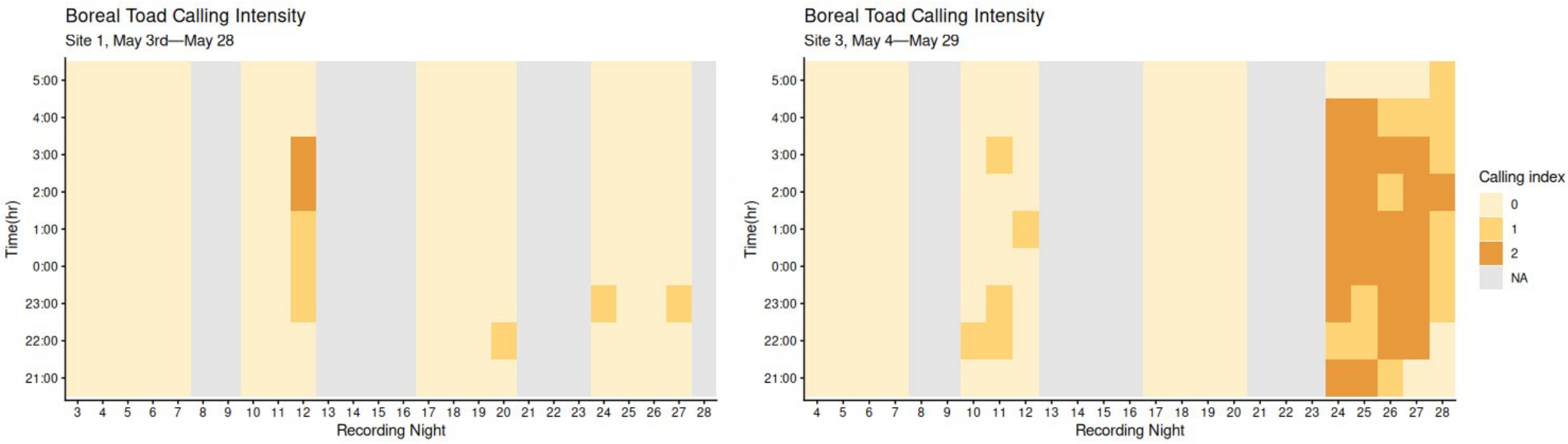
May month calling activity of Boreal Chorus Frogs(*Pseudacris maculata*) from southeast Lois Hole Centennial Provincial Park. Both recorders were active from 2100 MST to 0500 MST per night. Site 1(left) was active from May 3rd to May 28, Site 3(right) was active from May 4th to May 29th. Calling indexes were: "0" for no activity, "1" for fragmented non-overlapping calls, "2" for continuous overlapping calls, and "NA" to indicate no monitoring.

Figures 6-7 – Wood Frog Calling Intensity



May month calling activity of Wood Frogs(*Lithobates sylvaticus*) from southeast Lois Hole Centennial Provincial Park. Both recorders were active from 2100 MST to 0500 MST per night. Site 1(left) was active from May 3rd to May 28, Site 3(right) was active from May 4th to May 29th. Calling indexes were: "0" for no activity, "1" for fragmented non-overlapping calls, "2" for continuous overlapping calls, and "NA" to indicate no monitoring.

Figures 8-9 – Boreal Toad Calling Intensity



May month calling activity of Boreal Toads(*Anaxyrus boreas*) from southeast Lois Hole Centennial Provincial Park. Both recorders were active from 2100 MST to 0500 MST per night. Site 1(left) was active from May 3rd to May 28, Site 3(right) was active from May 4th to May 29th. Calling indexes were: "0" for no activity, "1" for fragmented non-overlapping calls, "2" for continuous overlapping calls, and "NA" to indicate no monitoring.

Discussion and Recommendations

This study successfully confirms the presence of *A. boreas*, *L. sylvaticus*, and *P. maculata* in the southeast parcel. Based on site activity, both P.maculata and L. sylvaticus breed in the riparian stretch of the parcel. Compressed cattails situated more inland provide females suspended structures to lay egg masses while standing cattails form a barrier for fish, preventing predation of eggs and early-stage tadpoles. Wood Frog choruses are complete by the first week whereas Boreal Chorus Frogs persist through May, likely into June. The discovery of Boreal Toads is especially significant: While individuals were heard near site 1, only one calling group was detected consistently in site 3. Terrain features that aid in A. boreas breeding include ground depressions with accumulated water. These microsites are the result of uprooted or fully decomposed trees. These sites can also serve as cool and humid areas to replenish moisture during post-reproduction movements. Wood frogs and Boreal Chorus Frogs likely breed in the John E. Poole wetland as well(figure 3).

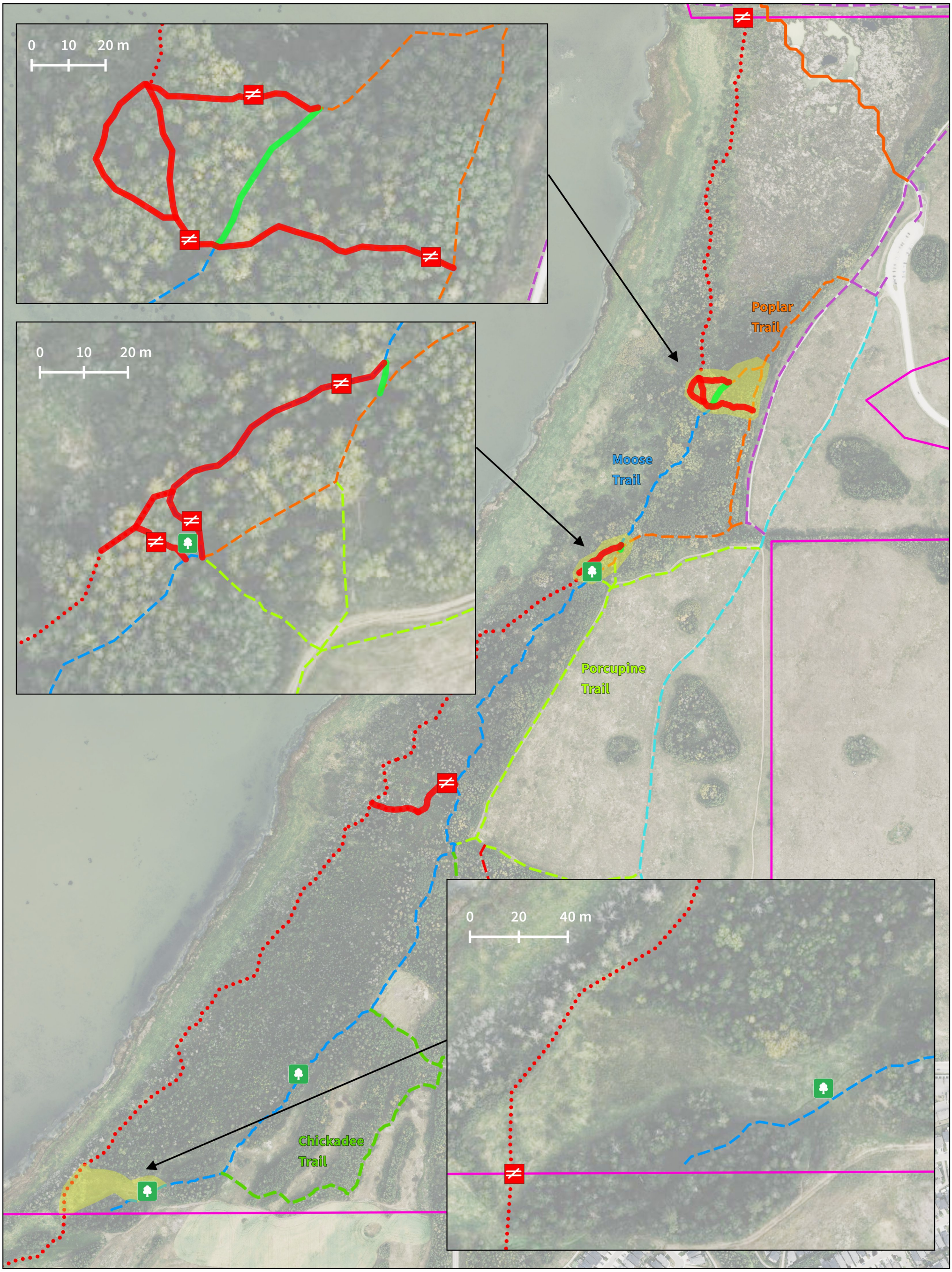
Internal and External Risks:

When comparing the three detected species, both L. sylvaticus and P. maculata appear to be better off: both species are actively breeding and possess large calling groups. P. maculata and L. sylvaticus breeding habitats are likely more widespread as this riparian habitat extends further along the south bank(figure 3).
A. boreas appears to be worse off: breeding habitat is limited in the parcel, though it is possible that other suitable sites exist. Occupying the lowest frequency bands A. boreas vocal dominance is also the weakest, putting their calls at risk of being washed out by L. sylvaticus and/or ambient noise pollution, this could explain why calling activity occured later in May.

Informal trail use poses the most direct threat to each species as both physical disturbance and disease transmission can target species and/or associated habitat. While most visitors are deterred by high water levels, only a handful of determined individuals are needed to disrupt reproduction; mostly by trampling egg masses or individual anurans. Further habitat loss is possible over time from trail branching, braiding, and widening.
Winter use can also impact amphibian hibernation. While P. maculata and A. boreas hibernate in burrows and underground cavities below the frost line, L. sylvaticus hibernates in the subnivium zone, under leaf litter [5,1,6]. With enough snowfall, the snow pack forms an insulating layer, containing ground heat, and maintaining stable and warm temperatures below[11]. Informal trail use — from foot and cyclist traffic — compresses the snow, crushing hibernating amphibians or by thinning the snow pack, lowering its insulating properties, and exposing hibernating amphibians to much colder temperatures[3]. While climate ultimately influences snow depth, extensive compression can have immediate and widespread impacts.
Another trail-based risk is disease introduction. All three species — especially A. boreas — are susceptible to chytrid fungus. Given the parks proximity to two large cities and current informal trail access, the number of potential vectors are abundant. While direct amphibian mortality due to chyrid fungus has not been detected in Alberta, infections can compromise physiology and behavior, elevating mortality risk[2,10]. While watercraft can become vectors, the most direct source would be from informal trail users.

Based on these findings, we are recommending the following:

- **Close informal trail entrances within the subject area.** Closure signs and obstacles should be placed at the beginning of the following trails(figure 10). New trail openings should be marked temporarily with smaller signs. Consider placing brush or deadwood for continuous deterrence into the winter.
- **Simplify trail system.** The more decisions a visitor must make to navigate may predispose them to choose informal trails. By simplifying the current system, less effort is needed and minimizes accidental trail use (figure 10).
- **Place "sensitive habitat" signs.** Signs like this are already in key positions, three signs would be necessary at these sites(figure 10).
- **Increase ranger presence during May.** Presence of parks officials during busy periods can deter opportunistic visitors. Morning and afternoon walks are recommended(figure 10).
- **Continue monitoring efforts.** Continue monitoring efforts in western sections of the park. While general surveys should be conducted, mapping further Boreal Toad populations and habitat should be prioritized. Post-reproductive anuran movements should also be explored by predicting movements via connectivity models or through direct sampling if necessary[7].



Park Boundary
Trail Closure
Trail Realignment
Natural Barrier
"Sensitive Area" Sign

0 100 200 m

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Appendix

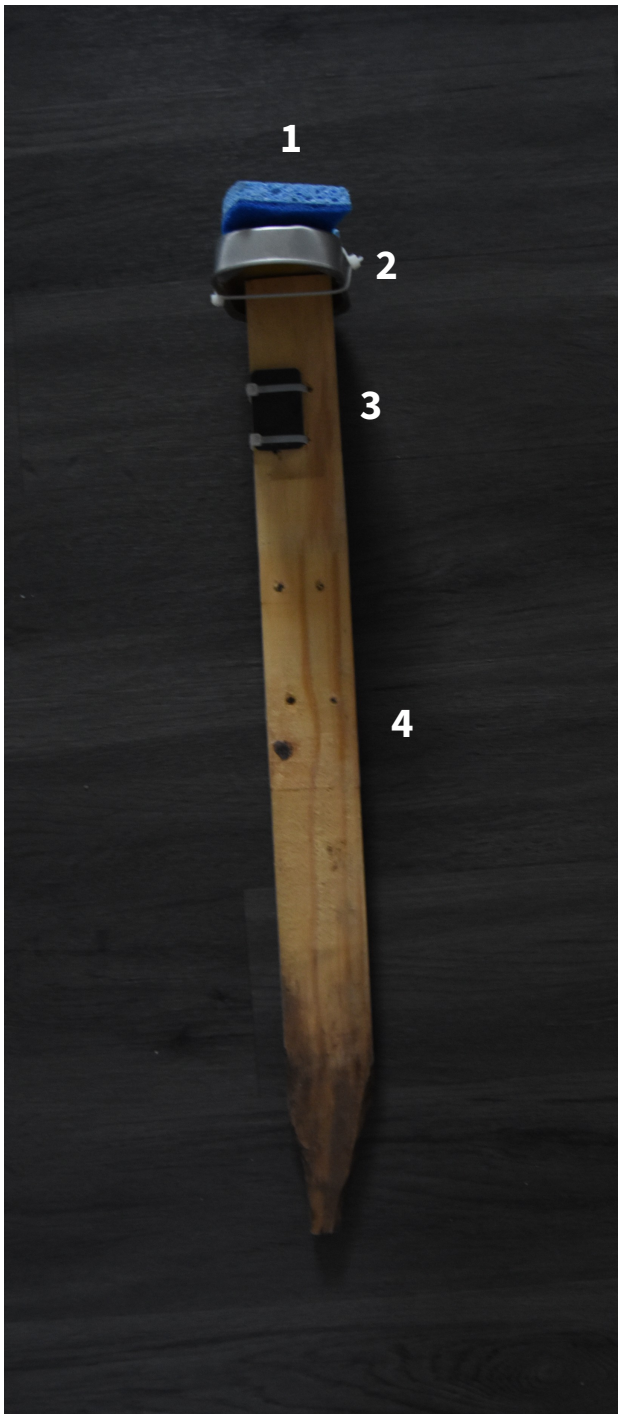
1 — Recording Nights:

Date	Night	Recording Status
May 3 – May 4	3	Site 2 discontinued — Recorder Inoperable
May 4 – May 5	4	
May 5 – May 6	5	
May 6 – May 7	6	
May 7 – May 8	7	
May 8 – May 9	8	Weekend, No Monitoring
May 9 – May 10	9	Weekend, No Monitoring
May 10 – May 11	10	
May 11 – May 12	11	
May 12 – May 13	12	
May 13 – May 14	13	Monitoring canceled — Dangerous weather
May 14 – May 15	14	Monitoring Canceled — Dangerous Weather
May 15 – May 16	15	Weekend, No Monitoring
May 16 – May 17	16	Weekend, No Monitoring
May 17 – May 18	17	
May 18 – May 19	18	
May 19 – May 20	19	
May 20 – May 21	20	
May 21 – May 22	21	Monitoring Canceled — Dangerous Weather
May 22 – May 23	22	Weekend, No Monitoring
May 23 – May 24	23	Weekend, No Monitoring
May 24 – May 25	24	
May 25 – May 26	25	
May 26 – May 27	26	
May 27 – May 28	27	
May 28 – May 29	28	Site 1 Monitoring Canceled — Wildlife Hazard

2 – Github repository link to project code and data file:

https://github.com/LWijesuriya/Amphibian-Survey-of-South-East-Lois-Hole-Centennial-Provincial-Park

3 — Recording Device:



5)

AMPHIBIAN AUDIO RECORDING UNIT

Permit #: 26-248

Property of Big Lake Environmental Support Society

For Additional Information, Contact: board@bless-env.ca

1

OVERNIGHT OPERATION ONLY

1 – Sponge
2 – Metal Rain Cover
3 – V-15 Audio Recorder
4 – Wooden Stake
5 – Identification Label(taped behind stake)